Name: ______________________

Numerical methods

Code: ______________________

Computer Science BSc

March 25, 2024

Group: ______________

Test 1

Solve the below problems! Write your solution of each problem on a separate page.

1. Consider the set of machine numbers $M = M(5, -3, 4)$. *(3+3+3+3 points)*

- **(a)** Identify $\varepsilon_0, \varepsilon_1$ and M_∞ in this set.
- **(b)** Calculate the machine numbers that correspond to 0.24 and 7.36.
- **(c)** Perform the machine addition $fl(0.24) \oplus fl(7.36)$.
- **(d)** Provide an absolute error bound for $fl(0.24)$, $fl(7.36)$ and their sum.

2. Calculate an absolute error bound for the value *(8 points)*

$$\frac{\sin(a+b)}{\sqrt{b+1}}$$

if $a = \pi/2$ and $b = 0$ are approximations with $\Delta_a = 1$ and $\Delta_b = 1/2$.

3. Let *(5+1+4 points)*

$$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 2 & 0 & -2 \\ 2 & 6 & 4 & -6 \\ 1 & 7 & 1 & -4 \end{bmatrix}, \quad b = \begin{bmatrix} -1 \\ 1 \\ 4 \\ 4 \end{bmatrix}.$$

- **(a)** Solve the system of linear equations $Ax = b$ using Gaussian elimination with **partial pivoting.**
- **(b)** Find $\det(A)$ from the previous results.
- **(c)** Give the LU decomposition of A. (Should not use the above results!)

4. Calculate $\mathrm{cond}_\infty(B)$ and $\mathrm{cond}_2(C)$ with respect to the matrices *(5+5 points)*

$$B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} -1 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 3 \end{bmatrix}.$$

5. Consider the polynomial *(5+2+3 points)*

$$P(x) = 3x^5 - x^4 + 2x - 1.$$

- **(a)** Give the Taylor polynomial of P centered at $a = 1$ using Horner's method.
- **(b)** Find the value of $P^{(4)}(1) + 2 \cdot P^{(2)}(1)$ based on the above.
- **(c)** Compute lower and upper bounds for the absolute values of the roots.

Good work!